Predictors and Underlying Causes of Medically Intractable Localization-Related Epilepsy in Childhood

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The goal of this study is to clarify the prognostic factors in childhood localization-related epilepsy in a tertiary medical center. Children (n = 113) with symptomatic and cryptogenic localization-related epilepsy were divided into groups of intractable patients (average seizure frequency: one or more per month during the 6 months before the last follow-up; n = 40) and well-controlled patients (no seizures for at least 1 year before the last follow-up; n = 73). Clinical and electroencephalogram (EEG) factors were examined to elucidate prognostic factors. The subtypes of epilepsies and causes were also investigated. Univariate analyses indicated that the following factors were correlated with seizure outcome: (1) seizure type at the first visit; (2) seizure frequency; (3) underlying cause; (4) age at onset of epilepsy; (5) status epilepticus occurring as the first seizure and before the first visit; and (6) diffuse epileptic discharges on first visit interictal EEGs. Multivariate analyses revealed that seizure type at the first visit, seizure frequency, status epilepticus before the first visit, and underlying causes were significant independent predictive factors. The rate of intractable patients was highest in multilobar epilepsy, followed by frontal-lobe epilepsy. Regarding etiologies, the intractable group contained nine patients with encephalitis of unknown origin and three each with localized cortical malformation and mesial temporal sclerosis.

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Introduction

In spite of advances in the treatment of epilepsy, a significant number of children with epilepsy do not respond well to antiepileptic drugs. Generalized epilepsies in childhood include many epileptic syndromes of which seizure outcome is unfavorable. On the other hand, a considerable number of patients with localization-related epilepsy also have intractable seizures. Compared with generalized epilepsy, seizure outcome is difficult to predict in childhood localization-related epilepsies. A significant number of children with medically intractable localization-related epilepsy may benefit from surgical treatment [1]. It is important to predict accurately those patients who can achieve acceptable seizure control within 3 or 4 years with the appropriate antiepileptic drug therapy in hospitals specializing in epilepsy and to select patients who may be candidates for epilepsy surgery.

Recently, new neuroimaging techniques have enabled the physician to observe subtle structural abnormalities in the brain. Clinical seizure types can be diagnosed with more accuracy by means of intensive monitoring of ictal scalp electroencephalograms (EEGs) with simultaneous video electromyogram-EEG recordings. Although brain abnormalities, especially mesial temporal sclerosis, are considered to be a major prognostic factor in intractable localization-related adult epilepsy [2], mesial temporal sclerosis rarely is diagnosed in childhood. Moreover, detailed investigations into the underlying causes of childhood localization-related epilepsy using the latest neuroimaging and EEG techniques have not yet been performed on a large number of children. To clarify the prognostic factors in childhood localization-related epilepsy treated in a tertiary medical center, a hospital-based study was performed.

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Table 1. Type of epilepsy and seizure control

Type of Epilepsy	Seizure Control: Well-controlled	Intractable	Others
Symptomatic (64)	26	32	6
Cryptogenic (59)	47	8	4
Total (123)	73*	40*	10

* Potential risk factors were analyzed in 113 patients.

Patients and Methods

The Okayama University Hospital is a tertiary medical center that receives patients with epilepsy for evaluation. Inclusion criteria for this study were as follows: (1) first visit to the center was from January 1, 1992 to December 31, 1994; (2) a follow-up with the center of 3 years or longer; (3) age at the first visit was less than 15 years; and (4) cryptogenic or symptomatic localization-related epilepsy. From these criteria, 123 patients were selected (Table 1). We analyzed the treatment effect on these 123 patients at the last follow-up. The last follow-up was defined as date of the last visit for each patient before May 31, 1998. The average period of follow-up was 4 years, 6 months.

We classified the state of seizure control into three types: *well-controlled*, defined as no seizures for at least 1 year before last follow-up; *intractable*, defined as an average seizure frequency of one or more per month during the 6 months before the last follow-up; and *others*. There were 73 well-controlled patients, 40 intractable patients, and 10 other patients. For the second step of this study, we excluded the 10 other patients in the analysis of risk factors related to unfavorable seizure outcome at the last follow-up. Subsequently, we analyzed risk factors in 113 patients with symptomatic or cryptogenic localization-related epilepsies (Table 1). In all intractable patients, seizures could not be controlled despite adequate trials of three or more antiepileptic drugs, used either alone or combination.

From the time of the first visit all patients were seen consecutively in the Okayama University Hospital by all authors except T.T. Reviewing medical records was performed by one of the authors (Y.O.), and information on the factors listed in Table 2 was gathered retrospectively.

Symptomatic epilepsies are considered the sequence of a known or suspected disorder of the central nervous system (CNS) [3]. We classified known and suspected disorders of the CNS into specific and nonspecific disorders. For specific CNS disorders, we detected definite focal abnormalities on magnetic resonance imaging (MRI) scans and known disorders of the CNS. We defined nonspecific causes as follows: (1) mental retardation and/or cerebral palsy with mild nonspecific MRI abnormalities, such as diffuse cortical atrophy or mental retardation and/or cerebral palsy without MRI abnormalities; or (2) only mild nonspecific MRI abnormalities. Mental retardation and cerebral palsy associated with specific CNS disorders were classified as specific CNS disorders.

Table 2. Factors examined

At or near the first visit
- Age at onset of epilepsy
- Sex
- Seizure types at the first visit
- Type of epilepsy at the first visit
- Seizure frequency at the first visit
- Status epilepticus as the first seizure
- Status epilepticus before the first visit
- History of febrile seizures
- History of neonatal seizures
- Family history of seizures within the second degree of relatives
- Etiology and underlying disorders
- Cranial MRI and/or CT scan
- EEG findings at the first visit (type of epileptic discharges)

At the last follow-up
- Follow-up duration
- Epileptic syndromes (subtypes of epilepsy)
- Antiepileptic drugs used

Abbreviations:
CT = Computed tomography
EEG = Electroencephalogram
MRI = Magnetic resonance imaging

In regards to subtypes of epilepsies, localization-related epilepsies were classified according to the International League Against Epilepsy (ILAE) classification into temporal, frontal, parietal, occipital lobe, and multilobar epilepsies [3]. When data were incongruent or insufficient, patients were classified as having localization-related epilepsy with undetermined localization. Clinical and EEG characteristics of four well-controlled patients fit well in those of atypical benign partial epilepsy of childhood that was proposed by Aicardi et al. [4]. At the first visit or during follow-up, subtypes of epilepsy were diagnosed based on the data gathered not only while the patient was seen as an outpatient but also while the patient was hospitalized.

We focused on epileptic discharges on EEGs at the first visit, and classified them as illustrated in Table 3. Patients who were admitted to our hospital underwent thorough EEG examinations, including simultaneous video electromyogram-EEG ictal recordings whenever possible and all-night sleep recordings were performed in some patients. Ictal EEGs using simultaneous video electromyogram-EEG recordings were available in 26 of 40 patients in the intractable group. They were also available in eight of 73 patients in the well-controlled group. As for neuroimaging examinations, MRI scans were performed in 96 (85%) of the 113 patients with symptomatic or cryptogenic localization-related epilepsy (54 of 58 patients with symptomatic localization-related epilepsy and 42 of 55 patients with cryptogenic localization-related epilepsy) using a Siemens Magnetom H15 unit (1.5 T; Siemens Medical Systems, Iselin, NJ). Another 17 patients underwent only computed tomography (CT) scan. MRI volumetric measurement of the mesial temporal structures was performed for patients with a suspicion of mesial temporal lobe epilepsy.

Statistical Analysis

Data were stored in an electronic database, and analysis of potential prognostic factors was based on all the information available at the time of the patient's first visit. These data were first assessed by examining the univariate relationship between each variable and seizure control.

After the univariate analysis we performed multivariate analysis to choose independently significant factors in univariate analysis. We discarded some factors in performing the multivariate analysis because of a zero cell.

Odds ratios and their confidence intervals were estimated by Cornfield's approximation with Epi Info 6-04 (US, the Centers for Disease Control and Prevention), and multiple logistic regression was performed with SPSS (SPSS, Inc., Chicago, IL).

Results

Characteristics of Symptomatic and Cryptogenic Localization-Related Epilepsy

The characteristics of 113 patients with symptomatic and cryptogenic localization-related epilepsy are illustrated in Table 3.

Table 3. Characteristics of 113 patients with symptomatic and cryptogenic localization-related epilepsy

	Intractable (40)	Well-controlled (73)
Sex (M/F)	1.5	1.2
Age at onset of epilepsy (years)		
1>	15	10
1≦, <3	6	22
3≦, <6	9	17
6≦	10	24
Family history of seizures (+)*	6	19
Neonatal seizures (+)	3	0
Febrile seizures (+)	3	9
Underlying disorders		
Specific	23	15
Nonspecific	9	11
Status epilepticus		
As the first seizure	8	5
Before the first visit	16	9
Seizure type at the first visit		
Complex partial seizures (+)	39	38
Seizure frequency at the first visit		
Daily or weekly	27	12
EEG at the first visit		
Single focus	9	29
Multiple foci†	12	20
Focal and diffuse epileptic discharges	18	19
No spikes	1	5
Subtypes of epilepsies		
ABPE	0	4
Temporal lobe epilepsy	8	11
Extratemporal lobe epilepsy	25	17
(Frontal lobe epilepsy)	(15)	(9)
(Occipital lobe epilepsy)	(6)	(8)
(Multilobar epilepsy)	(4)	(0)
Undetermined	7	41

* Family history of seizures indicates a history of seizures within the second degree of relatives; † Multiple foci mean two or more foci on EEGs.

Abbreviations:
ABPE = Atypical benign partial epilepsy of childhood
EEG = Electroencephalogram
F = Female
M = Male

Analysis of Factors Related to Seizure Outcome

Table 4 demonstrates the results of univariate analyses on the relationship between predictive risk factors and unfavorable seizure outcome at the last visit. Seizure type at the first visit, seizure frequency, underlying disorders, age at onset of epilepsy, status epilepticus occurring as the first seizure, status epilepticus occurring before the first visit, and diffuse epileptic discharges on interictal EEG at the first visit were correlated with the unfavorable seizure outcome.

Multivariate analyses were performed to assess the association between intractable epilepsy and potential predictive clinical and EEG factors. Seizure type at the first visit (odds ratio = 24.37, 95% confidence interval = 2.66, 223.29), seizure frequency (odds ratio = 5.96, 95% confidence interval = 1.80, 19.78), status epilepticus before the first visit (odds ratio = 5.56, 95% confidence interval = 1.30, 23.72), and underlying causes (specific and nonspecific causes or no causes; odds ratio = 5.35, 95% confidence interval = 1.59, 18.04) were independent predictive factors, although age at onset of epilepsy and interictal EEGs at the first visit were no longer significant in multivariate analyses.

Subtypes of Epilepsies

During follow-up many patients in the intractable group could be classified into specific subtypes of symptomatic and cryptogenic localization-related epilepsy because most of them were hospitalized and underwent intensive EEG monitoring. On the other hand, in many patients of the well-controlled group the interictal EEGs of the same patient revealed different foci on different EEG recordings and did not have a consistent epileptic focus during the clinical course. Moreover, clinical seizure manifestations were not particularly characteristic or revealing in many of these patients. Therefore we classify them as having localization-related epilepsy with undetermined localization of epileptic zones (Table 3). The rate of intractable patients was high in frontal lobe epilepsy (Table 3), and all four patients of multilobar epilepsy were intractable (Table 3).

Underlying Causes

In regards to underlying causes, 55 cryptogenic patients did not have definite underlying causes. In 58 symptomatic patients, 38 patients had specific causes and the other 20 had nonspecific causes. The relationship between specific underlying causes and seizure control is illustrated in Table 5. In the intractable group the postencephalitic state was most often observed, followed by mesial temporal sclerosis and localized cortical malformation.

Discussion

According to a multi-institutional study on intractable childhood epilepsy in Japan, approximately 13% of children with epilepsy experienced intractable seizures [5]. In the present study, one third of patients with cryptogenic and symptomatic localization-related epilepsy were classified as having intractable seizures at follow-up. The main reason for the high rate of intractable patients is that our university hospital is a tertiary referral center. Some of these patients with medically intractable epilepsy may be candidates for epilepsy surgery. Investigations into the prognostic factors and underlying causes of medically intractable epilepsy are important not only to find medically intractable and surgically treatable patients early in their clinical course but also to prevent toxicity from an overdose of antiepileptic drugs and useless polytherapy in medically intractable patients. The predictive factors elu-

Table 4. Univariate analysis of prognostic factors in symptomatic and cryptogenic localization-related epilepsy

Risk Factors	Intractable (40)	Well-controlled (73)	Odds Ratio	95% Confidence Interval
Type of seizures				
CPS (+)	39	38	35.92	4.68–275.53
CPS (−)	1	35		
Seizure frequency				
Daily or weekly	27	12	10.56	4.27–26.12
Less than weekly	13	61		
Underlying disorders				
Specific	23	15	9.01	3.34–24.3
Nonspecific	9	11	4.81	1.51–15.28
None	8	47		
Age at onset of epilepsy				
Less than 1 year	15	10	3.78	1.5–9.53
1 year or more	25	63		
Status epilepticus as the first seizure				
(+)	8	5	3.4	1.03–11.22
(−)	32	68		
Status epilepticus before the first visit				
(+)	16	9	4.74	1.85–12.16
(−)	24	64		
EEGs at the first visit				
Diffuse sp-w (+)	18	19	2.33	1.03–5.24
Focal sp/sp-w only or no spikes	22	54		

Abbreviations:
CPS = Complex partial seizure sp = Spike
EEG = Electroencephalogram sp-w = Spike-wave

cidated in this study will be useful in this decision-making process.

In most studies of the prognostic factors of epilepsy the authors analyzed clinical and EEG findings on all patients with all different types of epilepsy [5-8]. However, it is known that seizure outcome is related closely to type of epilepsy and epileptic syndromes. In hospital-based studies such as ours, first-step classification of types of epilepsy and epileptic syndromes can be performed in the early period after the first visit. It is therefore appropriate to analyze potential risk factors separately for patients in each group of first-step classification. According to multivariate analyses the presence of complex partial seizures, the frequency of seizures at the first visit, the underlying causes, and status epilepticus occurring before the first visit were significant independent prognostic factors. On the other hand, some factors, such as age at onset of epilepsy and EEG findings, were not significantly related to seizure outcome in the multivariate analyses. There is some controversy surrounding the prognostic factors of intractable epilepsy [6-8]. It is believed that differences in the data concerning prognostic factors are caused mainly by the differences between epidemiologic studies and hospital-based studies. In addition, analyses in many studies were performed on all patients with different types of epilepsy taken together.

Table 5. Seizure control and specific underlying causes

Underlying Causes	Intractable (23 Patients)	Well-controlled (15 Patients)
Encephalitis/encephalopathy	9	2
Localized cortical malformation	3	0
Mesial temporal sclerosis	3	0
Phakomatosis	2	2
Intracranial bleeding	2	0
Subdural hematoma	1	0
MELAS	1	0
Suspicion of chronic encephalitis	1	0
MRI abnormalities*	1	8
Others	0	3

* MRI abnormalities indicate high-intensity lesions of unknown cause on T_1- or T_2-weighted images in six patients, agenesis of corpus callosum in one, localized cortical atrophy of unknown cause in one, and asymmetric dilatation of ventricles of unknown cause in one.
Abbreviation:
MELAS = Mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke-like episodes

It is noteworthy that, with the exception of multilobar epilepsy, the rate of intractable patients was highest in frontal lobe epilepsy in the present study. In contrast, intractable patients are most often observed in temporal lobe epilepsy, such as mesial temporal lobe epilepsy in adult epilepsy [2]. Further investigation is needed to clarify whether this difference is caused by differences in age or in selection of patients.

Regarding underlying causes, the most characteristic finding of this study is that encephalitis was the most important underlying disorder of intractable symptomatic localization-related epilepsy in childhood. Encephalitis is becoming one of the major causes of intractable epilepsies in childhood in Japan [9-12]. In most patients, seizures begin at the acute stage of encephalitis and continue thereafter. In the acute stage, patients often experience repeat status epilepticus or frequent seizures and require general anesthesia with assisted respiration. Seizures in the acute stage often originate from several different regions in the individual patient, and these multifocal seizures continue in some patients after the acute stage. All of our patients with intractable postencephalitic epilepsy demonstrated the same typical findings as those previously reported [9-11]. These patients are often difficult to treat with medication and even epilepsy surgery.

It is well known that the most important cause of intractable localization-related epilepsy in adulthood is mesial temporal sclerosis, and mesial temporal lobe epilepsy is the best target for epilepsy surgery. In our study, three of eight intractable temporal lobe epilepsy patients were diagnosed as mesial temporal lobe epilepsy by means of MRI and MRI volumetric study. Like adult epilepsy, these pediatric patients are considered to be good candidates for epilepsy surgery [1,13]. In our study, there were some other intractable patients with focal brain lesions, such as localized cortical malformation. They are also possible candidates for epilepsy surgery, although epilepsy surgery is more challenging in these patients than those with mesial temporal lobe epilepsy [1,14,15].

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